

Lesson 01: The Plan for Differential Equations

Professor Leonard Differential Equations Master Series · Lesson 1

- **Direct Video Link:** [Lesson 01: The Plan for Differential Equations](#) · Duration: 3:17
- **Target Exam Scope:** 🎯 MIDTERM EXAM (Ch 1 — Course Philosophy & Overview)
- **Course Context:** Applied Ordinary Differential Equations (ENGR 213)

1. Whiteboard Lecture Intuition & Philosophical Framing

Professor Leonard introduces the Differential Equations master series by framing the fundamental transition from traditional calculus to the calculus of dynamic physical change. In ordinary calculus, one computes slopes and accumulated areas of known curves. In differential equations, one is provided with a relationship governing rates of change and must reconstruct the family of curves that obeys that physical law.

2. Core Theoretical Concepts & Mathematical Formulations

The Inverse Nature of Differential Equations

In standard Calculus I & II:

$$\text{Given a function: } y = f(x) \xrightarrow{\text{Differentiate}} \frac{dy}{dx} = f'(x)$$

In Differential Equations:

$$\text{Given a relationship: } F\left(x, y, \frac{dy}{dx}, \frac{d^2y}{dx^2}, \dots, \frac{d^ny}{dx^n}\right) = 0 \xrightarrow{\text{Integrate / Solve}} y = \phi(x)$$

Instead of seeking a static number $x \in \mathbb{R}$ that satisfies an algebraic polynomial, solving an ODE requires discovering a differentiable function $y = \phi(x)$ whose intrinsic rates of change satisfy the equation across an interval of existence I .

The 4 Pillars of the Concordia ENGR 213 Curriculum

1. Pillar 1: First-Order Analytical & Qualitative Methods:

- Explicit solution techniques: Separation of variables, linear integrating factor $\mu(x) = e^{\int P(x)dx}$, exact differentials with potential functions $\phi(x, y) = C$, and substitution transformations (Bernoulli, homogeneous $y = vx$, linear composition $u = ax + by + c$).
- Qualitative and geometric tools: Slope/direction fields, isoclines, solution curves, and Picard's Existence/Uniqueness theorem.

2. Pillar 2: Mathematical Modeling of Physical Systems:

- First-order balance laws: Conservation of mass in fluid tank mixtures ($\frac{dV}{dt} = \text{Rate}_{\text{in}} - \text{Rate}_{\text{out}}$), *Newton's law of cooling* ($\frac{dT}{dt} = -k(T - T_m)$), *Torricelli's draining tanks* ($A(h) \frac{dh}{dt} = -a c \sqrt{2gh}$), and *autonomous population models with carrying capacity* (the logistic equation $\frac{dP}{dt} = r P (1 - P/K)$).

3. Pillar 3: Higher-Order Linear Equations & Mechanical Oscillations:

- Homogeneous 2nd-order ODEs: The characteristic equation $ar^2 + br + c = 0$, distinct real roots, repeated roots (reduction of order $x e^{rx}$), and complex conjugate roots (Euler's formula generating oscillatory trigonometric bases $e^{\alpha x} \cos(\beta x)$ and $e^{\alpha x} \sin(\beta x)$).
- Non-homogeneous particular solutions: Method of Undetermined Coefficients (annihilators) and Variation of Parameters ($u_1 y_1 + u_2 y_2$ via the Wronskian determinant W).
- Mechanical and electrical systems: Mass-spring-damper systems ($m x'' + c x' + k x = F(t)$), transient vs. steady-state response, underdamped/overdamped/critically damped regimes, and pure resonance.

4. Pillar 4: Transform Methods & Linear Systems:

- The Laplace Transform: Operational calculus converting time-domain differential operators into frequency-domain algebraic fractions $\mathcal{L}y'' = s^2 Y(s) - s y(0) - y'(0)$, heaviside step functions, and impulse responses.
- Linear systems of ODEs: Vector-matrix representation $\mathbf{x}' = \mathbf{A}\mathbf{x}$, eigenvalue/eigenvector decompositions, and phase portrait classifications (stable/unstable nodes, saddle points, spiral sinks/sources).

3. Classroom Chalkboard Examples (Solved in Complete Baby Steps)

Problem 01.1: Chalkboard Problem 1.1: Distinguishing Algebraic Solutions from Function Families

Problem Statement:

Contrast the nature of the solution to the algebraic equation $x^2 - 7x + 12 = 0$ with the solution to the ordinary differential equation $\frac{dy}{dx} = 4y$. Detail the geometric interpretation of both.

Step-by-Step Whiteboard Solution:

- **Step 1: Solve the Algebraic Equation:**

$$x^2 - 7x + 12 = 0 \implies (x - 3)(x - 4) = 0 \implies x = 3 \quad \text{or} \quad x = 4$$

The solution consists of two discrete isolated points on the real axis $\mathbb{R}$: 3, 4.

- **Step 2: Solve the Differential Equation by Inspection / Separation:**

$$\frac{dy}{dx} = 4y$$

Divide by y (assuming $y \neq 0$) and integrate with respect to x :

$$\frac{1}{y} \frac{dy}{dx} = 4 \implies \int \frac{1}{y} dy = \int 4 dx$$

$$\ln |y| = 4x + C_1 \implies |y| = e^{4x+C_1} = e^{C_1} e^{4x}$$

Let $C = \pm e^{C_1}$ (and observe that $y = 0$ also satisfies $dy/dx = 0 = 4(0)$):

$$y(x) = Ce^{4x}, \quad C \in \mathbb{R}$$

- **Step 3: Geometric Comparison:**

- The algebraic equation yields discrete coordinates on a 1D number line.
- The differential equation yields a **one-parameter family of continuous curves** spanning the 2D Cartesian plane $\mathbb{R}^2$.
- For $C > 0$, curves lie entirely in the upper half-plane ($y > 0$) and exhibit exponential growth.
- For $C < 0$, curves lie entirely in the lower half-plane ($y < 0$) and grow negatively without bound.
- For $C = 0$, the curve collapses to the horizontal equilibrium axis $y(x) \equiv 0$.

[!WARNING] **Common Exam Pitfall:** Never treat the solution to a differential equation as a single number. It is fundamentally an infinite family of curves until boundary or initial conditions restrict C .

4. Key Takeaways & Exam Strategy Formula Card

- Watch the full whiteboard demonstration on YouTube: [Lesson 01: The Plan for Differential Equations](#)
- Master the classification and verification mechanics before attempting complex integrations.
- Always check domain restrictions and verify initial values back into the computed trajectory.